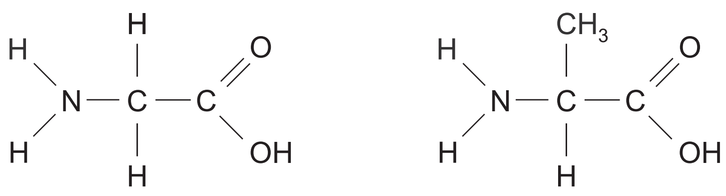
Examiner only

4. β-Bungarotoxin is a neurotoxin found in the venom of Krait snakes. The neurotoxin is a protein that causes muscle paralysis and eventual death.

The diagram below shows the structure of two amino acids found within the protein.

(a) (i) **Complete** the diagram, to show the products formed when these amino acid molecules are joined by a condensation reaction. [2]



(ii) State the bond formed. [1]

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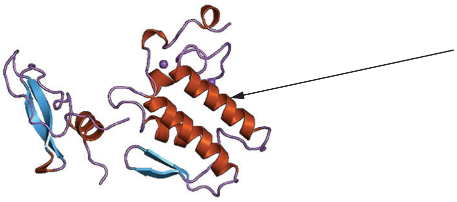
Examiner only

(b) The diagram below shows the structure of the β-bungarotoxin. The protein is composed of two polypeptide chains.



(i) State the highest level of protein structure exhibited by the β-bungarotoxin. [1]

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(ii) Name the structure labelled **X** and state how this structure is maintained. [2]

....

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(iii) State the minimum number of genes required to code for the β-bungarotoxin. Explain how you reached this conclusion. [1]

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....

(c) (i) One polypeptide chain of β-bungarotoxin contains 120 amino acids. State the minimum number of DNA nucleotides that would be required to code for this polypeptide. [1]

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(ii) Scientists have isolated, and sequenced, the gene for this polypeptide chain; it was found to contain over 2 000 nucleotides. Scientists have also isolated molecules of mRNA, which code for this polypeptide, and found that some mRNA molecules had a higher molecular mass than the others. What conclusions can be drawn about the production of this polypeptide? [5]

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